AGILE PROCESS MODEL

Introduction to Agile

What is Agile?

Agile is a flexible, iterative approach to software development.

Emphasizes collaboration, customer satisfaction, and adaptability.

Key Characteristics of Agile

- Iterative and Incremental development
- Customer collaboration over contract negotiation
- Embracing changes throughout the project
- Continuous delivery of working software
- Self-organizing teams

Agile vs. Traditional Waterfall Model

Aspect	Waterfall	Agile
Approach	Sequential	Iterative
Flexibility	Low	High
Customer Involvement	Low (at the end)	High (throughout)
Delivery Model	Single final delivery	Frequent small deliveries

- **Waterfall**: Sequential, rigid, phases like planning, design, development, testing, deployment.
- **Agile**: Iterative, flexible, continuous delivery, responsive to changes.

Agile Frameworks

Scrum: Roles (Scrum Master, Product Owner), Sprints, Daily Stand-ups.
Kanban: Visualizing workflow, limiting work in progress.
Extreme Programming (XP): Frequent releases, continuous testing, customer feedback.

Agile Process Overview

•**Steps in Agile Development**:

1. **Planning**: Identify goals and requirements.
2. **Designing**: High-level architecture and design.
3. **Developing**: Building features in short iterations (Sprints).
4. **Testing**: Continuous testing after each iteration.
5. **Reviewing**: Gathering feedback after each Sprint.
6. **Releasing**: Delivering shippable product increments.

Scrum Framework in Agile

•**Key Roles**:
•Scrum Master, Product Owner, Development Team
•**Key Artifacts**:
•Product Backlog, Sprint Backlog, Increment
•**Key Events**:
•Sprint, Daily Scrum, Sprint Review, Retrospective

Scrum Roles

The Scrum Team consist Product Owner, Scrum Master, and Development Team. Scrum Teams are self-organizing and cross-functional.

Product Owner

Responsible for maximizing the value of the product resulting from work of the Team

Scrum Master

Responsible for helping everyone in the team understand Scrum theory, practices, rules, and values

Development Team

Self-organizing & cross-functional group of people who do the hands-on work of developing and testing the product

Scrum Artifacts

Scrum defines three artifacts: Product Backlog, Sprint Backlog, and a potentially releasable product increment to help manage work.

Product Backlog

The Product Backlog is an ***ordered list of everything*** that is known to be ***needed in a product***

Sprint Backlog

Sprint Backlog is ***the set of Product Backlog items*** that the team commits to ***achieve in a given Sprint***

Product Increment

It is the ***sum of all the Product Backlog items completed during a Sprint*** and the value of the increments of all previous Sprints

Burndown Charts

Sprint burndowns are a ***graphical way of showing how much work is remaining in the sprint***, typically in terms of task hours

Scrum Events

Scrum defines four events (sometimes called ceremonies) that occur inside each Sprint: Sprint Planning, Daily Scrum, Sprint Review, and Sprint Retrospective

Sprint is a time-box of one month or less during which a useable, and potentially releasable product Increment is created

Sprint Planning

The entire Scrum Team collaborates and plans the work to be performed during a Sprint and defines the Sprint Goal

Daily Scrum

It is a every day 15-minute time-boxed event for the Development Team to inspect progress toward the Sprint Goal

Sprint Review

It is held at the end of the Sprint where the Scrum Team and stakeholders inspect the Increment and adapt the Product Backlog if needed

Sprint Retrospective

During a sprint retrospective the Scrum Team reflects upon how things went during the previous sprint and areas for improvement in the Sprint

DAILY SCRUM

- Is a short (15 minutes long) meeting, which is held every day before the Team starts working
- Participants: Scrum Master (which is the chairman), Scrum Team
- Every Team member should answer on 3 questions

SPRINT REVIEW MEETING

- Is held at the end of each Sprint
- Business functionality which was created during the Sprint is demonstrated to the Product Owner
- Informal, should not distract Team members of doing their work

PRODUCT BACKLOG

- Requirements for a system, expressed as a prioritized list of Backlog Items
- Is managed and owned by a Product Owner
- Spreadsheet (typically)
- Usually is created during the Sprint Planning Meeting
- Can be changed and re-prioritized before each PM

SPRINT BACKLOG

- A subset of Product Backlog Items, which define the work for a Sprint
- Is created ONLY by Team members
- Each Item has it’s own status
- Should be updated every day

SPRINT BURNDOWN CHARTS

- Are used to represent "work done".
- Are wonderful Information Radiators
- 3 Types:
  - Sprint Burn down Chart (progress of the Sprint)
  - Release Burn down Chart (progress of release)
  - Product Burn down chart (progress of the Product)

Case Study: Agile in Action

- A brief example of how Agile was implemented in a real-world project.
- Focus on collaboration, iteration, and flexibility.

Traditional VS Agile Model Working with Example

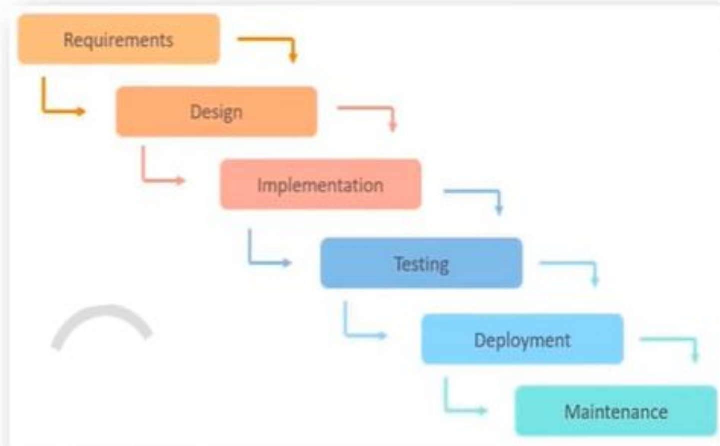
For Example:

Instagram Social Application:

Requirements are:

1. Follow – Unfollow Option
2. Edit profile
3. Search
4. Messaging
5. Post photos
6. Upload story
7. To make reels
8. Go Live

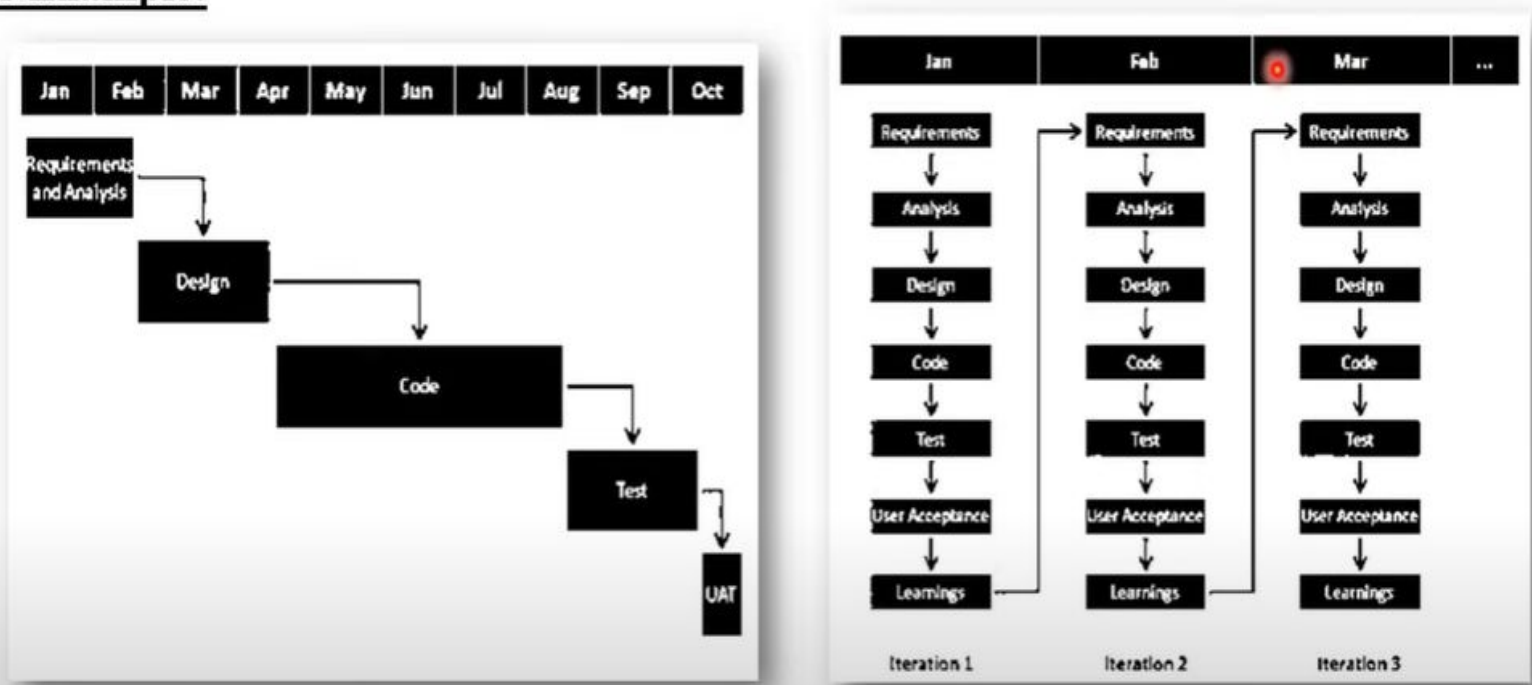
Waterfall Model



Agile Model

Traditional VS Agile Model Working with Example

<u>For Example:</u>

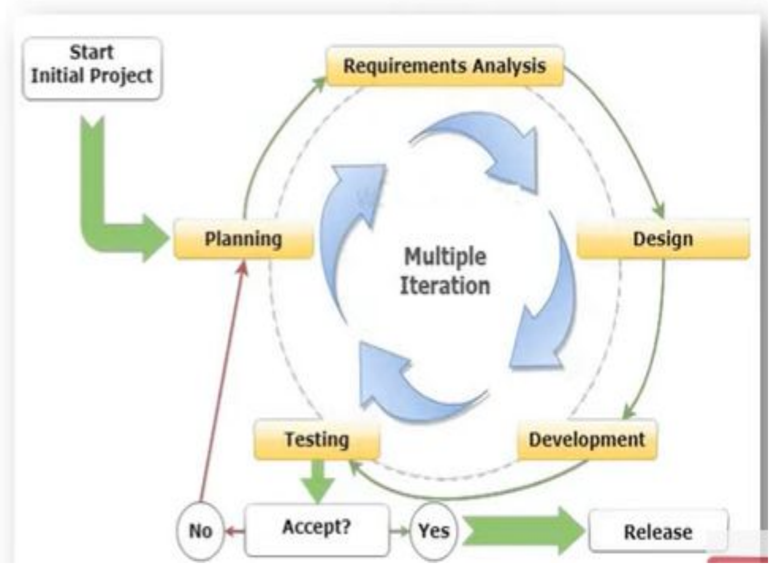


Waterfall Model
Time Span

Agile Model
Time Span

When to use the Agile Model?

1. When project size is large.
2. When frequent changes are required.
3. When a highly qualified and experienced team is available.
4. When a customer is ready to have a meeting with a software team all the time.
5. Projects with flexible timelines and budget



Agile Manifesto outlines 12 key principles

- The **Agile Manifesto** outlines **12 key principles** that guide Agile software development. These principles are focused on delivering value to customers, fostering collaboration, and enabling flexibility. Here are the **12 Agile principles**:

1. **Customer satisfaction through early and continuous delivery of valuable software:**
    1. The highest priority is to satisfy the customer by delivering working software early and continuously throughout the project.
2. **Welcome changing requirements, even late in development:**
    1. Agile processes harness change for the customer's competitive advantage, allowing the project to adapt to new requirements.
3. **Deliver working software frequently, with a preference for shorter timescales:**
    1. Teams should aim to deliver functional software in short, iterative cycles (such as every few weeks) to ensure continuous progress and feedback.

4. **Business people and developers must work together daily throughout the project:**

   Collaboration between technical and non-technical stakeholders is essential to ensure the project stays aligned with business goals.

5. **Build projects around motivated individuals:**

   Teams should consist of motivated individuals who are given the environment and support they need to get the job done, and be trusted to do their work effectively.

6. **Face-to-face communication is the most effective way to convey information:**

   The most efficient and effective method of communication is through in-person conversations within teams.

7. **Working software is the primary measure of progress:**

   The success of the project is determined by the delivery of functional, working software, not by the completion of documentation or following plans.

8. **Sustainable development is promoted:**

   Agile processes should promote a consistent pace of work that can be maintained indefinitely by developers, stakeholders, and users.

9. **Continuous attention to technical excellence and good design enhances agility:**

   Teams should strive for high technical quality and strong design practices, as this leads to better agility and long-term adaptability.

10. Simplicity—the art of maximizing the amount of work not done—is essential:

Teams should focus on simplicity and doing only what is necessary, avoiding unnecessary work and features.

11. Self-organizing teams produce the best results:

The best architectures, requirements, and designs emerge from self-organizing teams that are empowered to make decisions.

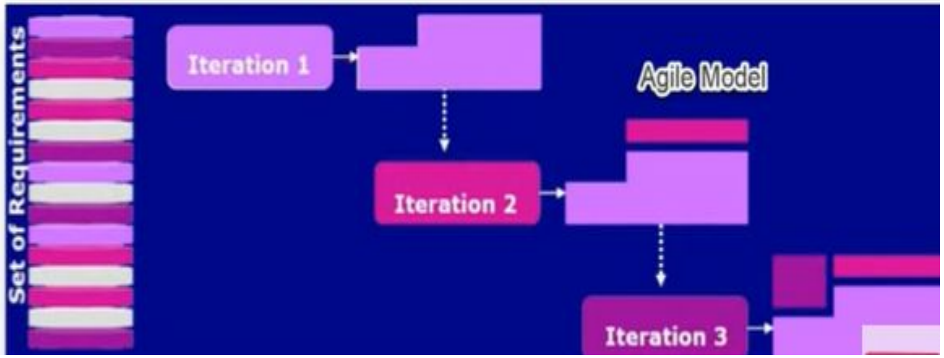
12.Regular reflection and adjustment to improve efficiency:

At regular intervals, the team reflects on how to become more effective and adjusts its behaviour and processes accordingly (e.g., through retrospectives).

These principles form the foundation of Agile development and promote continuous delivery, flexibility, and a strong focus on collaboration and customer needs.

Advantages of Agile Model

1. Supports customer involvement and customer satisfaction
2. Strong communication of the software team with the customer.
3. Little planning required
4. Efficient design and fulfils the business requirement.
5. Anytime changes are acceptable.
6. Provides a very realistic approach to software development
7. Updated versions of functioning software are released every week.
8. It reduces total development time.



Disadvantages of Agile Model

1. Due to the lack of proper documentation, once the project completes and the developers allotted to another project, maintenance of the finished project can become a difficulty.

2. Depends heavily on customer interaction, so if customer is not clear, team can be driven in the wrong direction.

Benefits of Agile Development

- Faster time to market
- Increased customer satisfaction
- Improved product quality
- Greater flexibility and adaptability
- Enhanced team collaboration

Challenges of Agile

- ❑Requires cultural shift
- ❑Managing customer expectations
- ❑Requires highly skilled teams
- ❑Scope creep if not managed effectively

Agile Tools

- ❑JIRA
- ❑Trello
- ❑Azure DevOps
- ❑Asana